

ANUBHAV SHRIJAN MISHRA

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Kattankulathur, Chennai, Tamil Nadu 603203

OBJECTIVE

Electronics and Communication Engineering undergraduate focused on VLSI, Embedded Systems, and Semiconductor Technologies with hands-on experience in Verilog HDL, UVM Verification, PCB Design, MEMS Research, and Hardware Development.

EDUCATION

SRM Institute of Science and Technology (SRMIST) B.Tech – Electronics and Communication Engineering (Data Science)	2023 – 2027 CGPA: 9.42 / 10
Chinmaya Vidyalaya, Bokaro Steel City, Jharkhand Class XII – PCM with Computer Science (CBSE)	2022 69.4%
Adiwasi High School, Sahibganj, Jharkhand Class X – Science (JAC)	2020 88.8%

PUBLICATION

Wearable Smart Face Mask Based on Functionalized Graphene for Real-Time Monitoring of Human Respiration

IEEE Sensor Journal · Co-author with Dr. Avik Sett · Nanotechnology Domain · [View Paper](#)

TECHNICAL SKILLS

- **HDL & Verification:** Verilog HDL, SystemVerilog, UVM, RTL/FSM Design, Functional Verification, Test-bench Development
- **EDA & Simulation Tools:** Icarus Verilog, GTKWave, ModelSim, Proteus, EasyEDA, SEMulator3D, Ansys Icepack, TinkerCad
- **MATLAB & Analysis:** MATLAB GUI (GUIDE/App Designer), MATLAB–Verilog Co-simulation, Waveform Analysis
- **Embedded Systems:** Embedded C, ATmega328, 8051, UART, SPI, I2C, Peripheral Interfacing, Arduino
- **Hardware & PCB Design:** PCB Design/Fabrication, Analog Circuits, Electronic Cooling, MEMS Fabrication
- **Programming Languages:** C, Python, SQL, R, Java, Kotlin, HTML/CSS/JS, LaTeX
- **Core Domains:** VLSI, Semiconductor Devices, IoT, Signal Processing, Data Analytics
- **Networking:** Junos OS, Routing, Switching, Cloud Networking

EXPERIENCE

- **Advanced Embedded System Developer Virtual Internship** — Microchip Technology Inc. Jan – Mar 2026
 - Completed a 10-week AICTE–EduSkills virtual internship on advanced PIC and AVR peripherals, ARM Cortex-M architecture, and MPLAB Harmony V3.
 - Developed bootloaders, CAN/CAN FD networks, and rapid-prototyping applications using Curiosity Nano and MPLAB X IDE
 - Achieved Grade O (Outstanding)
- **Structural Analysis Virtual Internship** — Ansys (part of Synopsys) Oct – Dec 2025
 - Completed a 10-week AICTE–EduSkills virtual internship on structural simulation and analysis fundamentals using Ansys tools

- Applied engineering analysis workflows to solve structural simulation problems
- Achieved Grade O (Outstanding)
- **Networking Virtual Internship** — Juniper Networks Jul – Sep 2024
 - Completed a 10-week virtual internship on core networking concepts supported by Juniper Networks’ Cloud & Automation Academy
 - Gained hands-on exposure to cloud automation practices and Juniper’s networking curriculum
 - Achieved Grade A (Very Good)

PROJECTS

Multiple Access Scheme for Wired Network — SRMIST 2026

ESP32 · W5500 Ethernet · CSMA/CD · UDP · Arduino

- Built a hardware CSMA/CD demonstrator using two ESP32 transmitter nodes and one ESP32 receiver node over W5500 Ethernet modules and a shared hub.
- Implemented carrier sensing, collision detection, jam signaling, random backoff, and retransmission, with real-time status shown on an OLED display.
- Designed the star-topology system architecture and algorithm flow, and documented protocol behavior across the Data Link Layer (IEEE 802.3).

LM386-Based Audio Amplifier Circuit using Solar Panel Mar 2026 – May 2026

SRMIST · Analog Circuit Design · EasyEDA

- Designed and tested a wireless audio transmission system using light as the carrier medium.
- Integrated solar panel as power source and verified signal integrity across the analog chain.

Micro Thermopile for Infrared Sensing — IISc Bangalore Feb 2026 – May 2026

MEMS Fabrication · SEMulator3D · LaTeX · Semiconductor Process Design

- Established a complete MEMS-based infrared thermopile sensor fabrication flow in collaboration with IISc Bangalore researchers.
- Designed and simulated device structure using SEMulator3D; documented deposition, lithography, and etching process steps.
- Gained hands-on exposure to semiconductor process optimization for sensing applications.

Wearable Nanomaterial-Based Smart Face Mask for Real-Time Respiration Monitoring Jan 2026 – May 2026

Nanotechnology · Functionalized Graphene · IEEE Published

- Developed a wearable sensor using functionalized graphene nanomaterials for continuous breath monitoring.
- Research outcomes contributed to an IEEE Sensor Journal publication on respiration monitoring.

Air Traffic Control System using Verilog HDL & MATLAB GUI 2025 – 2026

Verilog HDL · FSM · MATLAB (App Designer) · Icarus Verilog · GTKWave · VS Code

- Designed a priority-based FSM in Verilog HDL to handle 8 emergency request types with deterministic priority resolution.
- Integrated MATLAB GUI with Verilog via file I/O co-simulation for real-time landing priority output.

Digital Door Lock System Nov 2025

Proteus · ATmega328 · Embedded C

- Designed and simulated a secure digital door lock using ATmega328 microcontroller in Proteus.
- Implemented keypad-based access control with embedded C firmware and debounce logic.

One-Way Intercom Circuit using IC-386 May 2025

SRMIST · EasyEDA · TinkerCad · Analog Circuits

- Built and simulated a one-way intercom using IC-386 amplifier and validated before hardware build.

Home Security Alarm System using BJT and Logic Gates

Oct 2024

SRMIST · EasyEDA · PCB Fabrication

- Designed alarm circuit using BJT transistors and digital ICs with PCB fabrication in EasyEDA.
- Programmed trigger logic and validated operation on physical PCB.

CERTIFICATIONS

- **Computer Communication and Networks** — Udemy Aug 2026
- **Design Verification with SystemVerilog/UVM** — Udemy Oct 2025
- **8051 Microprocessor: Foundation of Microprocessors** — Intel Oct 2025
- **Verilog HDL Fundamentals** — Udemy Sep 2024
- **PCB Design and Fabrication** — Udemy Mar 2024

MEMBERSHIP

IEEE Electron Device Society (EDS) — SRMIST

May 2025 – Present

Senior Student Chair · Technical Member · IEEE Membership

- Led technical initiatives and coordinated 60+ member events, mentoring junior IEEE members and driving participation across the SRMIST EDS chapter.

VOLUNTEERING

Community Teaching Program — Saraswati Shishu Vidya Mandir, Jharkhand

Jun – Jul 2025

Mentored primary school students in academics as part of a community development initiative.

ACHIEVEMENTS

- 3rd Position — Best Project Exhibition, SRMIST (Jul 2024)
- Best Anchor Award — Saraswati Shishu Vidya Mandir (May 2021)

LANGUAGES

English (Professional)

Hindi (Native)

German (Basic)